AI & AGENT ENGINEERING ASSIGNMENT

FINANCIAL SUPPORT VOICE AGENT

A voice-first agent built on an existing TanStack Start + React console



FROM CONVERSATIONAL UI TO A REAL AGENT



REAL OPERATIONS

Understand financial-support requests; use real tools and APIs. Run read-only operations automatically.

SAFE BEHAVIOR

Ask for human approval before risky actions. Handle failures safely; answer in voice-friendly language.

Balance inquiry · card blocking · complaint submission · complaint tracking · live support availability

THE CONSOLE HAD A VOICE, BUT NO DOMAIN AGENT

ALREADY IN PLACE

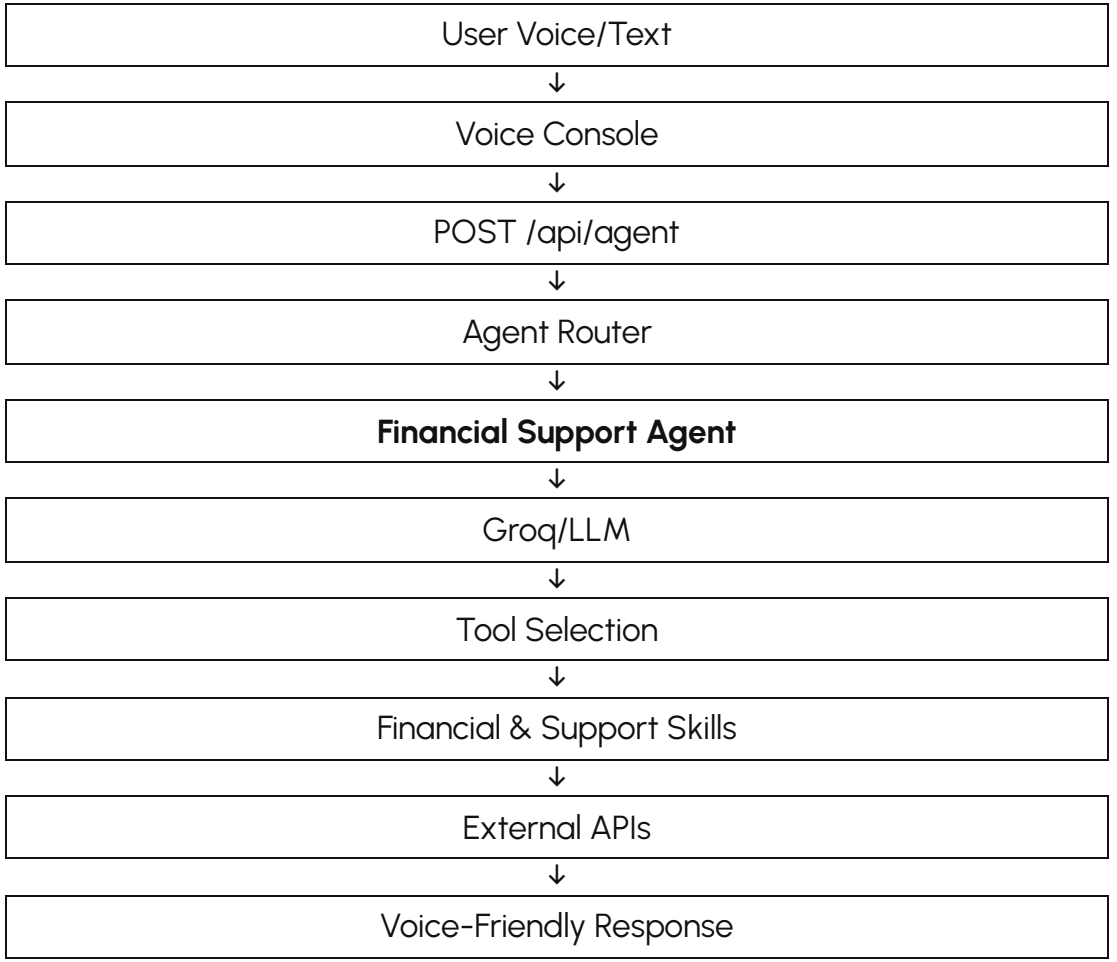
Voice input/output, hold-to-talk, hands-free mode, live execution steps, approval panel, 3D/WebXR interface, `/api/agent` bridge.

MISSING

A useful domain-specific agent behind it.

Risky actions

Tool Request → Action Proposal → Approve/Deny → Execute only after approval



CHANGE 1

A DEDICATED FINANCIAL SUPPORT AGENT

`openclaw/agents/financial-support/`

AGENT.TS

Orchestration, LLM calls, tool execution, approvals, final responses.

PROMPT.TS

Identity, supported tasks, safety rules, voice behavior.

TOOLS.TS

The functions available to the LLM.

CHANGES 2 & 3

FIVE REAL SKILLS ON A SHARED API CLIENT

READ-ONLY

Run immediately

- `get-account-balance`
- `follow-complaint`
- `is-live-chat-available`

STATE-CHANGING

Need explicit approval

- `stop-card`
- `submit-complaint`

`api-client.ts` provides GET/POST helpers, a 10-second timeout, basic auth, unified `ApiResponse`, and centralized errors.

Credentials stay server-side, never reaching the browser or LLM prompt.

CHANGE 4

EXPLICIT AGENT ROUTING

Requests could fall back to the original OpenClaw agent and answer about V3, P0, stages, `system_preflight`.

ROUTE

`src/routes/api/agent.ts`

FINANCIAL

`FINANCIAL_AGENT_IDS`
↓
`runFinancialAgent()`

OTHER AGENTS

↓
Default gateway

Financial requests use an isolated path and skip the project-management prompt.

CHANGE 5

RISKY ACTIONS WAIT FOR HUMAN APPROVAL

`STATE_CHANGING_TOOLS`

`finance.stop_card` · `support.submit_complaint`

01

PROPOSE

Return `action_proposal` instead of executing.

id, title, tool, summary, risk

02

REVIEW

Approval panel: approve or deny.

03

EXECUTE

The tool runs only after explicit approval.

CHANGE 6

FIXING THE LLM TOOL-CALLING PROTOCOL

Missing `tool_call_id` caused broken follow-ups and raw JSON in the UI.

BEFORE

```
{ role: "tool",
  content: result }
```

Tool result without tool_call_id

AFTER

```
{ role: "assistant", tool_calls }
{ role: "tool",
  tool_call_id: toolCall.id,
  content: result }
```

OpenAI/Groq format: assistant tool_calls, then a tool message with matching tool_call_id

Added a `withTools` flag to prevent accidental tool-call loops.

CHANGE 7

NORMALIZING VARIED LLM OUTPUT

`normalizeAgentResponse()`

Response variation	Mapping
Full JSON	Pass through
JSON without kind	Normalize
`{text}`	Convert to screen
Markdown-fenced JSON	Strip fence and parse
Plain Arabic text	Use as `screen.text`
Raw data	Replace with a readable reply

Resilient to normal variation without accepting garbage.

CHANGES 8 & 9

HONEST ERRORS AND SAFE DIAGNOSTICS

`humanizeToolError()` turns `UPSTREAM_TIMEOUT`, `NETWORK_ERROR`, `MISSING_INPUT`, and `MALFORMED_RESPONSE` into natural Arabic messages.

SERVER-SIDE LOGS

`api-client.ts`, `agent.ts`, and the route

- HTTP status and request path
- Unexpected finish reasons
- Failed follow-up calls

NEVER LOGGED

- API keys and passwords
- Full card numbers
- National IDs
- Auth secrets